Formulas set in the mathabx fonts.

$$x_i = \sqrt{1-\beta_i}\, x_{i-1} + \sqrt{\beta_i}\, z_{i-1} \tag{1}$$

Next, a radical over scripts.

$$x_i = x_{i-1} + \sqrt{\sigma_i^2 - \sigma_{i-1}^2}\, z_{i-1}$$

Relations and a sum with limits.

$$a \leqslant b < c, \qquad S = \sum_{i=1}^{N} \lambda_i \times \mu_i$$

And a few more symbols.

$$f(x) \propto \exp(-x^2) \pm 1$$